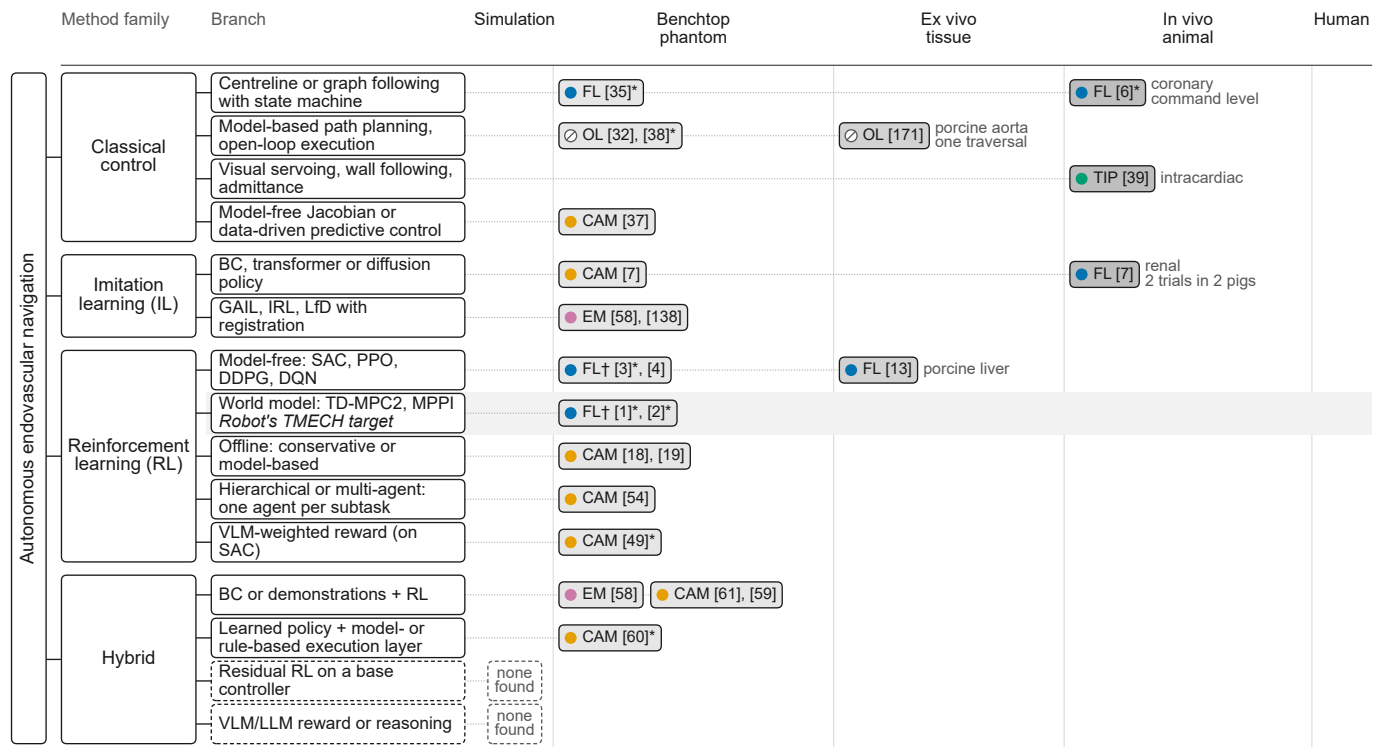

Method family
Branch
Simulation
Benchtop phantom
Ex vivo tissue
In vivo animal
Human
Autonomous endovascular navigation
Classical control
Centreline or graph following with state machine
Model-based path planning, open-loop execution
Visual servoing, wall following, admittance
Model-free Jacobian or data-driven predictive control
Imitation learning (IL)
BC, transformer or diffusion policy
GAIL, IRL, LfD with registration
Reinforcement learning (RL)
Model-free: SAC, PPO, DDPG, DQN
World model: TD-MPC2, MPPI Robot's TMECH target
Offline: conservative or model-based
Hierarchical or multi-agent: one agent per subtask
VLM-weighted reward (on SAC)
Hybrid
BC or demonstrations + RL
Learned policy + model- or rule-based execution layer
Residual RL on a base controller
VLM/LLM reward or reasoning
FL [35]*
OL [32], [38]*
CAM [37]
CAM [7]
EM [58], [138]
FL† [3]*, [4]
FL† [1]*, [2]*
CAM [18], [19]
CAM [54]
CAM [49]*
EM [58]
CAM [61], [59]
CAM [60]*
none found
none found
OL [171]
porcine aorta one traversal
FL [13]
porcine liver
FL [6]*
coronary command level
TIP [39]
intracardiac
FL [7]
renal 2 trials in 2 pigs
FL real fluoroscopy
CAM camera or simulated fluoroscopy
EM electromagnetic tracking
TIP tip camera
OL open loop
bench
ex vivo
in vivo
* field pending verification
† loop closed through the [5] tip tracker (accuracy never measured)
D1 verdict: under real fluoroscopy at bench stage or higher, no method family can be ranked (insufficient evidence to rank).